LogoLens: A Vision AI System to Measure Kit Sponsor Visibility and Value

Anonymous Author(s)
Affiliation withheld for double-blind review

***Abstract*—Sports clubs sell advertising space on their kits, yet published methods for measuring what that space delivers stop at aggregate screen time per brand. Two problems remain open: when the same sponsor appears on both teams' kits, boards, or officials, raw detection overcounts exposure unless each instance is attributed to its owner; and sponsors are priced by placement, yet no published system measures exposure at that granularity. We present *LogoLens*, an open, self-hosted vision pipeline a club can run on its own match video, with no enterprise broadcast integration, vendor contract, or in-house ML team. LogoLens contributes (i) a training-free *team attribution* method combining a zero-setup reference bootstrap, colour–embedding fusion, and track-level temporal voting, and (ii) *placement-level pricing*, attributing exposure to eighteen kit slots via pose estimation. On a production deployment at a professional club (identity withheld), detection reaches mAP@0.5 = 0.75 on a clip-disjoint split, versus an inflated 0.86 under the random-frame split common in this domain. Across eight of nine full-match videos, attribution excluded 41% of raw detections as opponent-owned, official-worn, or off-player, exposure otherwise priced as the sponsoring club's, and an audit of 184 track labels finds 92% correct; the ninth is reported as a failure case. Measured per-slot exposure is strongly non-uniform, the two shoulder slots alone carrying $\sim$30%, an empirical case for pricing placements individually. We report the calibration lessons and the audited failure case from taking the system to production.**

***Index Terms*—visual computing, applied machine learning, logo detection, team attribution, sports video analysis, sponsor visibility, media value estimation, open-source systems.**

I. Introduction

Sports clubs sell advertising space on their kits, but only a small number of clubs can verify what that space is actually worth. Our partner club is representative of the majority of the market this paper targets: a professional club that sells sponsor slots on its kit but, like most clubs outside the top handful of leagues, has no in-house analytics team and no enterprise sponsorship-measurement contract. (We withhold the club's name throughout this paper for commercial confidentiality; nothing in the results depends on the club's identity.)

A commercial layer already serves the sponsor-measurement need: Nielsen's sponsorship benchmarking [1], GumGum/Zoomph, Relo Metrics [2], and Shikenso all offer automated or semi-automated sponsor-visibility services. These are delivered as closed, hosted platforms: a club sends its footage to a third party and receives a report back, typically under a recurring contract and typically by integrating with an official broadcast feed. For a club with the budget and a broadcast deal this is reasonable, but it leaves three structural costs for everyone else: vendor lock-in and recurring fees; handing raw video and commercial data to an external party; and a dependency on broadcast access that many clubs outside the top leagues simply do not have, regardless of budget. Our framework instead is built around *self-service ingestion*: a club uploads its own match video, whatever it already has, such as its own match stream or publicly posted uploads, a livestream recording, or a fixed pitch-side camera, and the system produces exposure and EMV reports from that, with no dependency on an official broadcast feed or a vendor integration. We target clubs for whom these costs, not detection accuracy, are the binding constraint. By "accessible" we mean clubs with modest technical capacity but no analytics budget or vendor contract: self-hosting removes cost and lock-in, not the need to operate the system.

Even with a working, open-source logo detector, two problems remain unresolved by the closest published system, ExposureEngine [3] (Section II). First, the same sponsor frequently appears on both competing teams' kits, on perimeter advertising boards, or on match officials, so raw detection systematically overcounts a sponsor's exposure unless ownership is resolved per detected instance. Second, sponsors are priced by placement in practice (a chest-centre logo is worth more than one on a sock), yet no published system we are aware of attributes exposure, or price, at that granularity.

We present *LogoLens*, the system embodying these ideas, and make three contributions:

- **C1.** A formulation and training-free solution for *team attribution*: resolving which team owns a detected logo instance without training a separate model per opponent, by fusing a colour histogram with a general-purpose vision embedding and accumulating track-level temporal votes with hysteresis. Per-team references come from a three-tier bootstrap (club-supplied kit references $\rightarrow$ kit-anchor auto-clustering $\rightarrow$ luminance heuristic), so no per-match human configuration is required.
- **C2.** *Placement-level pricing*: attributing exposure, and therefore price, to eighteen individual kit slots via pose estimation, a granularity absent from both the manual AVE/EMV literature and recent AI-driven valuation systems (Section II).
- **C3.** An open, self-hosted reference architecture built around *self-service video ingestion* rather than a broadcast-feed integration, together with the production lessons of deploying it at a real club: a two-pass sampled/full-fps trade-off, graceful degradation under

partial failure, two calibration corrections (visibility floor, train/test split leakage) that changed reported numbers substantially, and an audited team-attribution failure case with an operational mitigation.

C1 and C2 are the primary technical claims; C3 is the system contribution that makes them deployable, together with the evidence of how they behave in a real deployment.

II. Related Work

A. Logo detection

Logo detection is a comparatively mature problem [4]; we build on a standard fine-tuned open-weight detector, not a new detection architecture, and treat detection accuracy as a means to an end, not a contribution. Public sports-video benchmarks such as SoccerNet [5] target broadcast footage and carry no sponsor-ownership or kit-placement annotations, so they cannot evaluate the two problems this paper addresses; our evaluation therefore uses the deployment's own annotated footage.

B. Team classification and jersey-based re-identification

Assigning a detected player to a team from jersey appearance is an active area in its own right, including joint re-identification, team-affiliation, and role classification for sports tracking [6], jersey-number recognition under uncertainty [7], and earlier work combining spatial constellation features with jersey-number recognition for player identification [8]; see also the broader computer-vision- in-sports survey [9]. Most of this literature performs per-frame colour or embedding clustering for team assignment. We build directly on this and add a track-level temporal voting scheme with hysteresis, together with a zero-manual-setup reference bootstrap, the differentiation is in *how* team identity is made stable and cheap to configure per match, not in team classification itself.

C. Sponsor-visibility and ad-surface systems

A separate, older line of work localizes static advertising surfaces in broadcast video to insert or replace advertising content [10]–[12], a related but different problem from ours, which measures the exposure of sponsors already physically on screen. Closer to our goal, the closest published system is ExposureEngine [3], which introduces an oriented-bounding-box detector for sponsor logos and a visibility/exposure analytics pipeline for soccer broadcasts, reporting mAP@0.5 of 0.859 on a purpose-built Swedish top-flight soccer dataset drawn from official broadcast coverage. As far as its published description shows, it does not address logos shared between opposing teams or officials, does not attribute exposure below brand level, does not discuss deployment cost, and (like the commercial vendors below) assumes access to an official broadcast feed, not club-captured video. This last point matters in practice: most clubs outside the top leagues have no broadcast coverage to draw on at all, independent of budget. Commercial vendors (Nielsen [1], GumGum/Zoomph, Relo Metrics [2], Shikenso) confirm that the underlying problem is commercially important, but their methods are closed, unpublished, and not independently reproducible.

D. Existing pricing and valuation models, manual and AI-driven

The industry-standard manual approach is Advertising/Earned Media Value Equivalency (AVE/EMV): convert the size, duration, and placement of exposure into the cost of an equivalent slot of paid advertising, typically via a hand-set placement multiplier. Our own Tier-3 pricing formula (Section IV-C) is drawn from exactly this family of formulas, as used in prior industry and academic practice. The AVE methodology is also the target of sustained, public criticism from industry bodies: AMEC, IPR, PRSA, PRCA, and ICCO have each issued position statements rejecting AVE as unreliable [13], citing arbitrary multipliers, no signal for content or audience sentiment, and treating every viewer as identical regardless of placement or audience quality. Our placement-level pricing (C2) makes one specific weakness of this criticism (the arbitrariness of a single, flat multiplier) more principled, by grounding a portion of the multiplier in measured, per-slot exposure share. It does *not* address the deeper criticism that AVE-style metrics ignore audience quality, sentiment, or brand recall, a scope limitation we return to in Section VI.

On the AI-driven side, recent systems apply machine learning directly to sponsorship valuation: SSPAIN.ai [14], a survival-analysis model trained on more than 5,800 historical sponsorships to forecast sponsorship renewal probability and duration, and Relo Metrics' [2] combined computer-vision-and-machine-learning pipeline for automated media value and dynamic pricing. Both are methodologically related to more general machine-learning-based dynamic pricing, e.g. in e-commerce [15], which reacts to real-time demand signals rather than sports-specific placement data. None of this AI-driven work, as far as published material shows, prices exposure below the level of a brand's aggregate on-screen time, and none discusses low-cost or self-hosted deployment.

In summary, no published work we are aware of simultaneously addresses (a) team/opponent-overlap attribution, (b) placement-level pricing, in either the manual or AI-driven valuation literature, and (c) deployment accessibility for clubs without an enterprise budget or an in-house ML team. This is the gap this paper targets.

III. System Architecture

LogoLens is organized as a single orchestrated job per uploaded video (Fig. 1). Two design choices are central to the accessibility claim. First, every optional pipeline stage degrades gracefully: if a stage's dependency is unavailable (e.g. no team reference could be bootstrapped), the stage is skipped and logged rather than failing the job, so a club always gets a usable, if partial, result. Second, infrastructure is abstracted behind swappable interfaces: the default configuration uses local file storage, an embedded database, and an in-process job queue, with an opt-in path to object storage, a managed database, and a distributed queue for clubs whose volume justifies it, so a deployment starts at near-zero cost and scales only when needed.

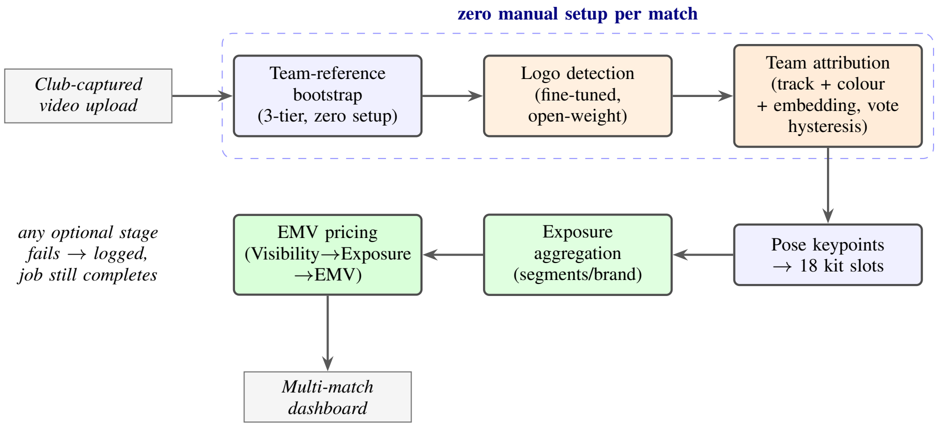

Fig. 1. **End-to-end pipeline.** An orchestrated job runs frame sampling, a zero-manual-setup team-reference bootstrap, logo detection, team attribution, pose-based kit-slot assignment, exposure aggregation, and EMV pricing, feeding a multi-match dashboard. Every optional stage degrades gracefully on failure rather than aborting the job, and infrastructure (storage, database, job queue) sits behind swappable interfaces so a deployment can start at near-zero cost.

The pipeline also separates two passes over the video: a sparsely sampled pass drives the exposure/EMV numbers at a fraction of the compute cost, and a full-frame-rate pass produces a smooth annotated preview for review. This trade-off is reported (Section V) rather than hidden inside one uniformly-sampled pass.

IV. Method

A. Detection and annotation

The logo detector is a YOLO26m model fine-tuned at 1280-pixel input on manually annotated frames of the same club-captured video the framework runs on in deployment, which is potentially more variable in camera stability, framing, and quality than the broadcast footage used to train prior published detectors. Annotation frames are not sampled uniformly: a selection pipeline keeps only in-play frames (players present, pitch dominant), ranks them by jersey-region sharpness (Laplacian/Tenengrad/FFT), and removes near-duplicates by perceptual hash, so human effort goes to informative frames. We then use a model-assisted labelling loop, deliberately built on off-the-shelf tooling (Roboflow Label Assist) so a club without ML staff can reproduce it: a small hand-labelled seed set (tens of images per class) trains an assist model that proposes boxes on the remaining frames for a human to correct, before the final detector is trained on the combined set. This does not eliminate manual annotation, the main residual barrier to full accessibility (Section VII); we make no claim of an annotation-free pipeline. Per-frame detection sits behind a swappable backend interface, so a transformer-based detector (RF-DETR) can replace the YOLO backend without touching the rest of the pipeline; all detection numbers reported in this paper use the YOLO26m backend. In deployment, detections below a confidence floor of 0.20 are discarded, but confidence is otherwise used as a continuous weight in the visibility score (Section IV-C), not a gate, a deliberately recall-first choice: in our deployment data, detections with confidence below 0.4 account for 29% of pose-attributed detections but only 9.5% of quality-weighted exposure, so genuine small or motion-blurred kit logos are retained at little cost to the aggregate, and spurious ones are further suppressed by the minimum-duration rule of Section IV-C.

B. Team attribution

Detected persons are tracked across frames; for each track, a jersey crop is classified by fusing a colour histogram with a general-purpose vision embedding, and classifications are accumulated per track with hysteresis, so that a single blurred or occluded frame cannot flip an already-confident team label. Reference examples for each kit are obtained automatically rather than by manual setup: from official kit images when available, and otherwise by clustering the video's own early frames and selecting the cluster closest to a simple luminance heuristic (e.g. a known dark alternate kit); Fig. 2 illustrates the cascade and the resulting per-team references.

Each detected logo instance is then attributed to the team of its nearest tracked person. When the evidence for opponent ownership is insufficient (a track has not yet accumulated enough confident votes), the instance is *kept*, not dropped, a revenue-safe design choice: an under-confident drop would silently understate a paying sponsor's measured exposure, a worse failure mode than a few false positives that a human reviewer can catch.

C. Visibility, exposure, and EMV pricing

Exposure and pricing are computed in three tiers, adapted from the manual AVE/EMV family discussed in Section II-D rather than proposed as a new pricing theory: (1) a per-frame *visibility* score combining detection size, on-screen position, and detector confidence,

$$v \;=\; \min\!\Big(1,\; \sqrt{\tfrac{A_{\text{box}}}{A_{\text{frame}}}} \;\cdot\; e^{-d_c^2/(0.3\,W)^2} \;\cdot\; c\Big), \qquad (1)$$

where $A_{\text{box}}/A_{\text{frame}}$ is the box-to-frame area ratio (square-rooted so a huge logo cannot dominate), d_c the distance of the box centre from the frame centre, W the frame width,

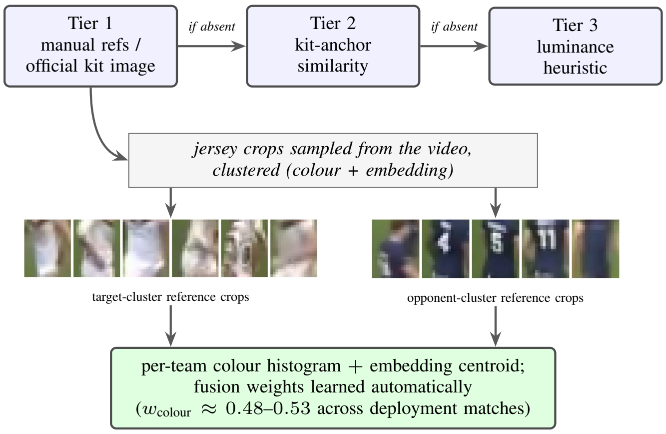

Fig. 2. **Zero-manual-setup team-reference bootstrap (part of C1).** A three-tier cascade selects which cluster of jersey crops is the target team; the chosen clusters become per-team references (colour histogram + vision-embedding centroid) with fusion weights learned automatically. Crops shown are an illustrative reconstruction on one daylight fixture using the same detect–crop–cluster procedure; they are torso-only and deliberately pixelated, since the deployed system stores only the resulting feature centroids, never images.

and c the detector confidence; (2) temporal aggregation of visibility into per-brand *exposure* segments, consecutive same-track detections with v above the floor are grouped, gaps longer than 2.5× the sampling interval split a segment, and segments shorter than 0.5 s are discarded as likely spurious; and (3) a conversion to *estimated media value* using an audience size, a cost-per-thousand (CPM) baseline, and a placement multiplier supplied per deployment (the implementation additionally supports sponsor-category and prime-time multipliers, both disabled in the deployment reported here). A specific calibration lesson from deployment: the visibility floor of 0.1 suggested in the pricing methodology we adapt, applied as published, discards nearly all real, small, off-centre sponsor logos typical of kit sponsorship, under our size term such logos score in the 0.03–0.08 range, so we use an empirically set floor of 0.02. A sensitivity sweep over the 13,439 pose-attributed detections of the eight audited deployment matches (Section V) makes the stakes concrete: raising the visibility floor from 0.02 to 0.05 would delete 71% of quality-weighted exposure, and applying the published 0.1 floor would delete 98% of it, whereas raising the detector-confidence floor from 0.20 to 0.6 removes under 5%. The visibility floor, not detector confidence, is the operative gate, and small changes to it dominate every other calibration choice in the pipeline.

Placement-level pricing. Each attributed detection is additionally assigned to one of eighteen kit slots (e.g. chest-centre, sleeve, sock) via pose-estimation keypoints, so that exposure, and hence the placement multiplier applied in Tier 3, can be reported and priced per slot rather than only per brand (the measured per-slot distribution from our deployment is shown in Fig. 5). This directly targets the "arbitrary multiplier" criticism of AVE-style pricing (Section II-D) by grounding part of the multiplier in measured, slot-specific exposure share, while leaving audience-quality and sentiment signals as future

TABLE I
EFFECT OF THE TRAIN/VALIDATION SPLIT POLICY ON MEASURED DETECTION ACCURACY (FINE-TUNED YOLO26M, BEST VALIDATION EPOCH). THE RANDOM-FRAME FIGURE IS INFLATED BY NEAR-DUPLICATE FRAMES LEAKING ACROSS THE SPLIT, NOT BY A BETTER MODEL.

Split policy	mAP@0.5	mAP@[.5:.95]
Random-frame	0.86	0.57
Clip-disjoint, same data	0.70	0.42
Clip-disjoint, extended data	**0.75**	**0.44**

work (Section VI).

V. EXPERIMENTS

A. Setup

We evaluate on our partner club's own match video (identity withheld), taken from the club's publicly available match uploads, not an official top-tier broadcast feed. The annotated detection dataset covers seventeen sponsor classes and is severely imbalanced, from roughly 200 instances for the most frequent kit sponsor down to single digits for the rarest, a property of real kit sponsorship, not of our sampling, since a chest sponsor simply appears in almost every frame while a shorts-leg sponsor does not. Detector training and evaluation use a clip-disjoint split: all frames from the same clip or match are kept entirely within one split. This choice matters more than any architectural one we made: because analytics frames are sampled at 2 fps, neighbouring frames are near-duplicates, and a naive random-frame split leaks them across train and validation. The same YOLO26m detector scores mAP@0.5 = 0.86 under a random-frame split but 0.70 under the clip-disjoint split of the same source data (Table I), the higher number reflects leakage, not a better model, and published results in this domain rarely state which split strategy they used. One scope note: in the nine full-match runs below, per-team references came from tier 1 of the bootstrap cascade, club-supplied kit references built once per kit, not per match; the automatic tiers have been exercised on other uploads in this deployment but are not separately evaluated here, so the "zero per-match setup" property of C1 carries tier-1 evidence only in this paper.

B. Detection accuracy

On the initial annotated set, the clip-disjoint validation score is mAP@0.5 = 0.70 (mAP@[.5:.95] = 0.42); after extending the dataset with additional annotated clips under the same clip-aware split policy, the current detector reaches mAP@0.5 = 0.75 (mAP@[.5:.95] = 0.44, precision 0.65, recall 0.74). Rare-class AP is unreliable at these instance counts and the validation split intentionally holds no instances of the rarest classes, so aggregate mAP should be read together with the per-class breakdown.

Three observations frame whether 0.75 is "enough." First, the operating domain is harder than professionally produced broadcast: Fig. 3 spans the confidence spectrum the detector produces on this footage, most exposure comes from clean,

Fig. 3. **The detection-confidence spectrum in club-provided match video** (boxes; confidence under each crop) from one night and one daylight fixture. Top row: representative high-confidence detections (0.87–0.97), sharp, large, frontal marks; in deployment this regime carries 64% of quality-weighted exposure. Lower rows: the hard tail, floodlit night play, heavy motion blur, sub-0.1%-of-frame scale, and fold/occlusion deformation, routine in a lower-tier club stream. Marks in the hard tail are inherently illegible at capture, which is precisely why a hard confidence gate would delete genuine exposure; the pipeline keeps them and lets visibility weighting and the minimum-duration rule (Section IV-C) suppress spurious ones.

high-confidence detections, but a hard tail of small, motion-blurred, deformed marks is routine in club-captured video and drags aggregate mAP in a way that clean broadcast benchmarks never see. Second, the number is not comparable to published results unless the split protocol is stated: our own detector reaches 0.86 when near-duplicate frames leak across the split (Table I), so a protocol-silent comparison against, e.g., ExposureEngine's 0.859 [3], reported on broadcast footage in a different domain, would be meaningless in either direction. Third, per-frame mAP is not the quantity a sponsor report is built from. Exposure is computed from temporally aggregated segments (Section IV-C): the gap tolerance bridges isolated per-frame misses within a segment, and the minimum-duration rule discards isolated false positives, so moderate per-frame accuracy on a tracked, repeatedly visible logo can still yield accurate exposure-seconds. The operative accuracy metric for the system's end output is therefore the exposure-seconds error against manual timing (Section V, EMV validation), with mAP reported for completeness and future comparability. The per-class picture reinforces this. On the clip-disjoint model, the normalized confusion matrix is almost purely diagonal apart from its background row: the detector essentially never confuses one sponsor for another (inter-class confusion is negligible), and almost all of its error is missed detections of small or rare logos falling to background, concentrated in the single-digit-instance classes that the split cannot both train and validate. For an exposure system this is the benign error mode: a missed frame on a repeatedly visible logo is bridged by temporal aggregation, whereas a brand-for-brand swap (which we do not observe) would misattribute revenue. A held-out clip-disjoint *test* split, distinct from the validation split used here for model selection, remains future work.

C. Team attribution

We audited the per-track attribution overlays of all nine deployment matches: three frames sampled across each match's overlay window, and every *decided* track label in them judged against the kit visible in the frame, 184 track-label observations in total, each traceable to an archived frame (the per-observation log accompanies the paper). Overall, 91.8% of decided labels are correct: 90.7% of 86 target labels and 92.9% of 98 opponent labels. The errors are informative. Half of the eight incorrect target labels sit on match officials or sideline stewards in high-visibility clothing, a third appearance class the binary reference set does not model; the rest are opponent kits. All seven incorrect opponent labels assign the club's *own* players to the opponent class, the revenue-critical direction, and concentrate in the first seconds after kick-off, before temporal votes stabilise, and in the one fixture where both teams wore dark kits. Notably, decided-label accuracy is roughly uniform across capture conditions; difficult conditions (night, long camera range) manifest instead as tracks that never accumulate a confident label and logos that never attach to a track, as *missing* decisions rather than wrong ones, which the revenue-safe keep-if-unknown policy of Section IV-B is designed to absorb. These observations cover the native-frame-rate overlay windows ($\sim$35 s per match); end-to-end validation of the full-match 2 fps analytics pass is subsumed by the exposure comparison of the EMV validation below.

At the deployment scale, across nine full-match highlight videos processed by the pipeline, team attribution flagged and excluded 44% of raw logo detections (11,161 of 25,153) as owned by the opposing team or match officials, or as not attached to any tracked player (e.g. perimeter advertising boards). We did not take this aggregate at face value: the match with the highest exclusion rate (78%) cannot be validated by the audit of Section V-C. Its overlay window shows largely correct decided labels, but also close-up scenes in which the club's own white kit is assigned to the opponent class (despite reference features a luminance check confirms are correct), and its long-range night feed leaves many detections unattached to any track, so the audit can neither confirm nor localise what drove the full-match rate. We therefore treat that match's measurement as untrustworthy, report it as a failure case (Section VI), and exclude it from the headline figure: over the remaining eight matches, attribution excluded 41% of raw detections (9,383 of 22,863), with per-match rates of 21–50% on dark-kit fixtures and 26–72% on white-kit fixtures, the upper end driven by the noisy night-time captures above, in which many distant tracks never accumulate a confident label. Without attribution, every excluded detection would have been counted, and priced, as target-club kit exposure. These figures quantify how much raw exposure the filter removes in production, not yet its per-instance accuracy, which is the subject of the labelled-clip evaluation of the previous subsection. A finer counterfactual (running the pipeline with attribution enabled and disabled on clips where a specific sponsor appears on both teams' kits, to price the exact doublecounting that attribution removes for that sponsor) is a natural next step that the current deployment does not yet isolate; the 41% aggregate above is the deployment-scale evidence we report here.

D. Exposure, placement, and sampling error

Fig. 4 shows the per-sponsor exposure segments the pipeline recovers for one full highlight video, which is the intermediate representation all exposure and EMV figures are computed from. Beyond confirming end-to-end operation, the sorted totals expose a commercially meaningful fact: on the same kit in the same match, delivered exposure spans a 27× range (110 s for the top sponsor to 4 s for the bottom one), driven largely by which slot each sponsor occupies (Fig. 5), evidence that selling kit space at a flat rate misprices most of it. We have quantified the internal validation, the cost of the 2 fps sampling rate against native-rate detection (above). The external one, comparing system exposure-seconds against an *independent manual timing* of logo appearances for a full match, remains open; it is the single most important measurement before this system's exposure figures can be called independently verified. EMV itself is a downstream consequence of exposure, not an independently verifiable quantity: CPM, audience size, and placement multiplier are deployment-supplied business inputs, not measurable ground truth, so multiplying by them is arithmetic, not a claim to validate against an external reference, unless an independent buyer valuation is available for direct comparison.

To isolate what the 2 fps analytics pass itself costs, independent of detector error, we additionally ran the detector on every frame (native 50 fps) of a contiguous three-minute in-play chunk of one deployment match, and computed brand-level exposure with identical segment rules at both rates. The sampled pass does not under-measure; it *over*-measures: +63% summed across brands relative to the native-rate reference, ranging from +18% for the brand with the most sustained visibility to several times over for brands whose visibility is fragmented into sub-second bursts. Two mechanisms account for this: an isolated 2 fps sample is credited a full 0.5 s quantum, and the gap-bridging window scales with the sampling interval (1.25 s at 2 fps), merging appearances that the native-rate measurement correctly separates. Duration weighting already discounts sub-second segments in the EMV computation, so quality-weighted exposure is less affected than raw seconds; nonetheless, raw exposure-seconds from the sampled pass should be read as upper-bound-leaning, most so for marginal sponsors, and the comparison against independent manual timing above remains the definitive check.

Fig. 5 reports the measured per-slot exposure distribution that placement-level pricing (C2) is built on. Two observations matter for pricing. First, the distribution is far from uniform: shoulder and chest-adjacent slots dominate attributed exposure, while sleeve, sock, and back-centre slots individually carry under 2% each, supporting the practice of pricing slots differently, but from measurement rather than convention. Second, the ordering is not the conventional price ordering: in this deployment the shorts-back slot outranks chest-centre, an artefact of how the club's own match camera views play, and precisely the kind of deployment-specific evidence a flat industry multiplier cannot anticipate.

E. Accessibility evidence

End-to-end processing throughput is measured directly from the deployment's job log: the nine full-match videos of Section V-C (87.7 video-minutes in total) were each processed from upload to finished report (including the 2 fps analytics pass, the full-frame-rate annotated preview, the team-attribution overlay video, pose-based slot attribution, and EMV computation) in 88.4 wall-clock minutes in total, i.e. ≈1.0× real time (per-match range 0.81–1.12×), on a single consumer workstation (NVIDIA RTX 5060 Ti 16 GB, Intel Core i5-13400F, 16 GB RAM). A deliberate cost split underlies this: the only step that needs more GPU than the deployment box is detector fine-tuning. Our training logs make the boundary concrete: the small initial set trained in 6.4 h of wall-clock time, but the extended set would take over a week on consumer-class hardware, so fine-tuning is done on rented cloud GPUs (Google Colab) instead. That is an *occasional* cost, paid at initial setup or when onboarding a new sponsor class; every *recurring*, per-match cost runs on the owned consumer workstation above, in about the video's own duration. Renting the training and owning the inference is the cost structure that makes self-hosting viable for a club without an infrastructure budget. The annotation cost is bounded by the modest scale the approach requires: the detector reported here was trained on roughly 1,100 boxes across 17 sponsor classes, seeded from tens of images per class and grown through the model-assisted loop of Section IV rather than by exhaustive manual labelling. We did not instrument per-class wall-clock annotation time in this deployment, and report the dataset scale in its place; a controlled measurement of annotation effort per sponsor class is a concrete item of future work and would sharpen the accessibility claim.

VI. Discussion and Limitations

Team attribution assumes a stable kit within a match; a mid-season kit change requires a reference refresh. Colour-and-embedding fusion for team classification has not yet been stress-tested across a wide range of lighting conditions, and this is not hypothetical: in one night fixture, the audited overlay shows the club's own white kit assigned to the opponent class in close-up scenes despite correct stored reference features, and that match's full-pass exclusion rate (78%) could not be validated (Section V-C), most of its genuine sponsor exposure was likely deleted silently. The audit also exposes a modelling gap: match officials and sideline stewards in high-visibility clothing account for half of the incorrect *target* labels (Section V-C), suggesting the binary target/opponent reference set should grow a third, explicit officials class. Because the dominant failure direction is revenue-critical, a deployment should monitor the per-match exclusion rate and treat an outlier (here, 78% against a fleet median near

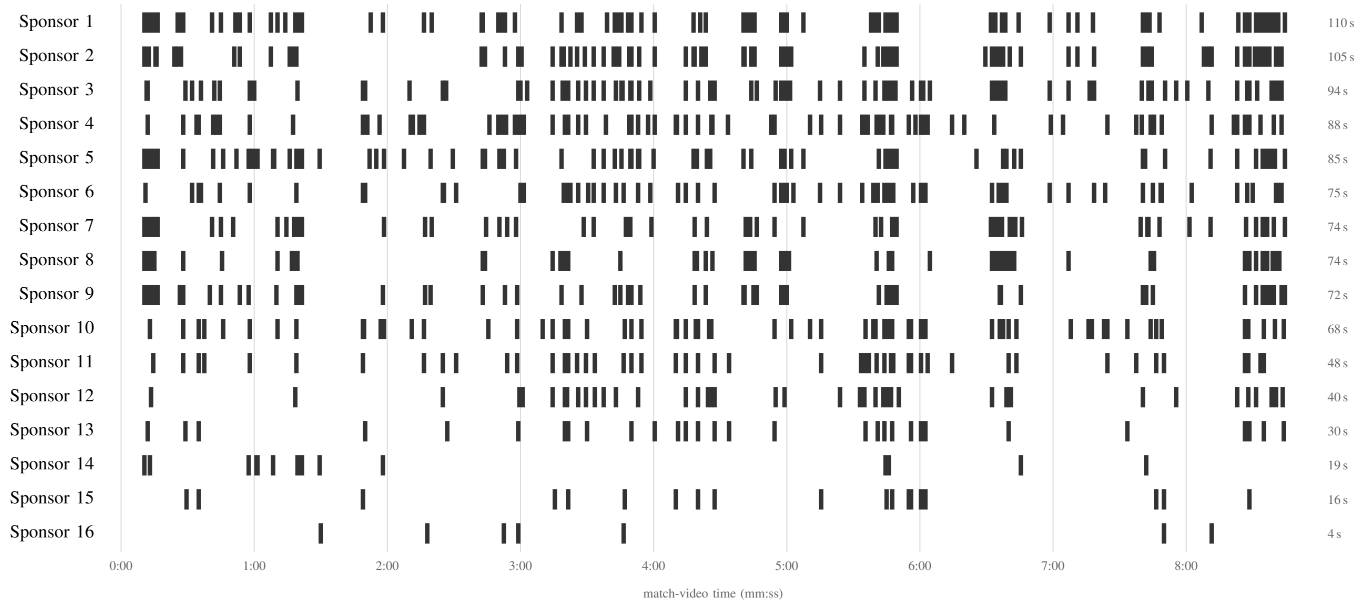

Fig. 4. **Per-sponsor exposure timeline produced by the deployed pipeline** for one club-captured highlight video (8.9 min, black-kit fixture, 2 fps analytics pass, team attribution enabled; this match's attribution overlay was visually audited and assigns target/opponent correctly, Section V-C). Each bar marks an interval in which the sponsor was detected on the target team's kit after attribution; sponsors are sorted by total on-screen time (right column) and numbered rather than named, since the full sponsor list would identify the club. Very short intervals are widened to a minimum drawable width for legibility; totals are computed from the underlying segments, not from the drawing.

40%) as a signal for human review, not a valid measurement. Targeting a club's own lower-tier match video also means tolerating a wider range of camera stability, framing, and production quality than top-tier broadcast systems; we have not yet characterized robustness across capture sources (a produced club stream vs. a fixed pitch-side camera), and flag it as an open question. The sparse sampling rate used for the analytics pass trades exposure-time precision for processing cost, at a measured +63% over-measurement (Section V) concentrated on marginal, flicker-exposed brands. Since this bias is sponsor-favourable, a deployment should disclose it to sponsors rather than quietly benefit. Manual annotation, while reduced by the model-assisted loop, remains a real, non-eliminated cost when onboarding a new sponsor class, and is the main residual barrier to the accessibility claim in this paper (Section VII). As discussed in Section II-D, our placement-level pricing addresses one specific, well-documented weakness of AVE-style valuation (arbitrary multipliers) but not the broader critique that such metrics do not capture audience quality, sentiment, or brand-recall impact; we do not claim to have solved sponsorship valuation, only to have made one component of it less arbitrary and more transparent.

Data and ethics statement. The footage used in this study is publicly available match video, obtained from YouTube and provided to us by our partner club; it depicts professional players during public fixtures and involves no new collection of personal data. The system nonetheless processes images of real people, and constrains what it does with them by design: players are tracked only as transient, anonymous instances for kit-slot attribution, with no facial recognition and no jersey-number or player identification in the deployed pipeline, and its output is brand- and media-value analytics, not player-performance or biometric data. All figures in this paper withhold the club's identifying marks (name, crest, scoreboards, broadcast watermarks). A club deploying the framework should apply the player-media consent terms already governing its own footage.

VII. Future Work

The main remaining barrier to full accessibility is the manual annotation effort required per sponsor class. We are exploring an annotation-free direction that exploits a structural property of team sports: a club wears one fixed kit for a season and advertising boards are static, so each distinct logo-bearing surface could in principle be identified once, at its clearest occurrence, and propagated to every other by tracking and geometry rather than repeated visual classification. Early results are encouraging but preliminary, and are not part of this paper's claims. Other directions include synthetic data for rare viewing conditions and an audience-quality-aware model for converting exposure into media value, addressing the deeper AVE critique of Section VI.

VIII. Conclusion

We presented LogoLens, an open, self-hosted system for sponsor exposure and media-value measurement that addresses two problems left open by prior published work: attribution of logos shared between competing teams, and pricing at the level of individual kit placements. The evidence comes from a production deployment at a professional club: clip-disjoint detection accuracy reported alongside the inflated random-split

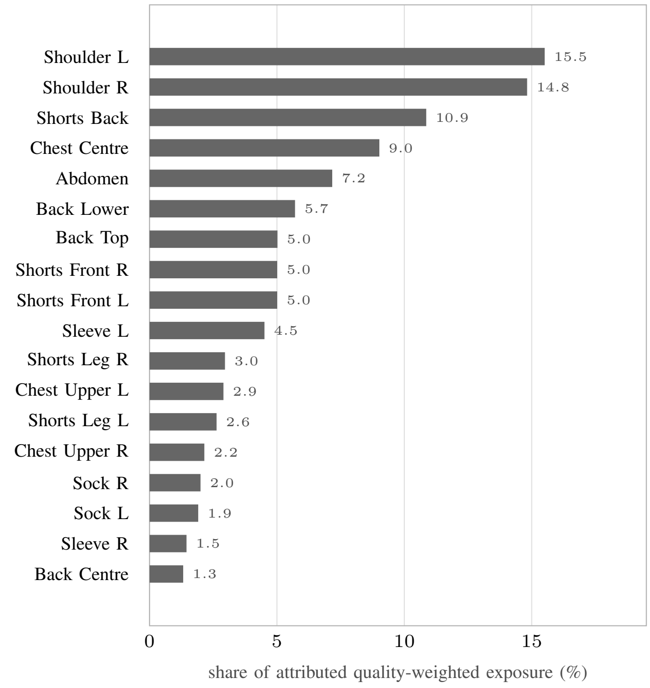

Fig. 5. **Where kit exposure actually accrues: share of quality-weighted exposure per kit slot**, aggregated over eight full-highlight analyses (Section V-C; the ninth match is excluded after the attribution audit described there, since its kept detections are not trustworthy). Percentages are weighted by each match's attributed exposure and cover only detections the pose stage could assign to a slot. The distribution is strongly non-uniform: the two shoulder slots alone carry $\sim$30% of attributed exposure, an order of magnitude more than a sock slot, exactly the structure a single flat placement multiplier cannot express.

figure it would otherwise be mistaken for (0.75 vs. 0.86); a deployment-scale measurement of the overcounting attribution prevents (41% of raw detections across eight matches); a per-slot exposure distribution that quantifies, for the first time in published work, how unevenly a kit actually delivers its sponsors' visibility; and an audited failure case with an operational mitigation. We position the pricing methodology against both the manual AVE/EMV tradition and recent AI-driven valuation systems, and are clear about which part of the well-documented criticism of media-value-equivalency metrics it does, and does not, address. Two measurements remain before the exposure story is complete: per-instance attribution accuracy, and exposure error against manual timing (Section V).

References

[1] Nielsen, "Sponsorship media value benchmarking report," Nielsen, Tech. Rep., 2025, industry report, not peer-reviewed.

[2] Relo Metrics, "Relo Metrics: Ai-powered sponsorship analytics with computer vision and multi-modal AI," Company press material, 2025, industry system, not peer-reviewed.

[3] M. H. Sarkhoosh, F. Øye, H. N. Sørlie, N. H. Vu, D. Johansen, C. Midoglu, T. Kupka, and P. Halvorsen, "ExposureEngine: Oriented logo detection and sponsor visibility analytics in sports broadcasts," in *Proc. IEEE International Symposium on Multimedia (ISM)*, 2025, also available as arXiv:2510.04739.

[4] S. Hou, J. Li, W. Min, Q. Hou, Y. Zhao, Y. Zheng, and S. Jiang, "Deep learning for logo detection: A survey," *ACM Transactions on Multimedia Computing, Communications, and Applications*, vol. 20, no. 3, pp. 72:1–72:23, 2023, also available as arXiv:2210.04399.

[5] A. Deliège, A. Cioppa, S. Giancola, M. J. Seikavandi, J. V. Dueholm, K. Nasrollahi, B. Ghanem, T. B. Moeslund, and M. Van Droogenbroeck, "SoccerNet-v2: A dataset and benchmarks for holistic understanding of broadcast soccer videos," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW, CVSports)*, 2021, also available as arXiv:2011.13367.

[6] A. M. Mansourian, V. Somers, C. De Vleeschouwer, and S. Kasaei, "Multi-task learning for joint re-identification, team affiliation, and role classification for sports visual tracking," in *Proc. 6th International Workshop on Multimedia Content Analysis in Sports (MMSports)*, 2023, also available as arXiv:2401.09942.

[7] Ł. Grad, "Single-stage uncertainty-aware jersey number recognition in soccer," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW, CVSports)*, 2025, pp. 6148–6156.

[8] S. Gerke, A. Linnemann, and K. Müller, "Soccer player recognition using spatial constellation features and jersey number recognition," *Computer Vision and Image Understanding*, vol. 159, pp. 105–115, 2017.

[9] B. T. Naik, M. F. Hashmi, and N. D. Bokde, "A comprehensive review of computer vision in sports: Open issues, future trends and research directions," *Applied Sciences*, vol. 12, no. 9, p. 4429, 2022.

[10] C. Xu, K. W. Wan, S. H. Bui, and Q. Tian, "Implanting virtual advertisement into broadcast soccer video," in *Advances in Multimedia Information Processing — PCM 2004, Part II, LNCS 3332*. Springer, 2004.

[11] H. Shah and S. Chaudhuri, "Automated billboard insertion in video," in *Computer Vision — ACCV 2007, Part I*. Springer, 2007.

[12] C. Li, G. Cai, and L. Chen, "Billboard advertisement detection in sport TV," in *Proc. 7th International Symposium on Signal Processing and Its Applications (ISSPA)*, 2003, hAL id hal-01588349.

[13] AMEC (International Association for the Measurement and Evaluation of Communication), "Barcelona principles 3.0," https://amecorg.com/barcelona-principles-3-0/, 2020, principle 5: "AVEs are not the value of communication".

[14] J. A. Jensen, "SSPAIN.ai: Sport sponsorship predictive artificial intelligence network," Texas A&M Stories, https://stories.tamu.edu/news/2026/01/06/ai-tool-uses-data-to-redefine-sports-sponsorships/, 2026, press coverage; no peer-reviewed publication located as of July 2026.

[15] M. Nowak and M. Pawłowska-Nowak, "Dynamic pricing method in the e-commerce industry using machine learning," *Applied Sciences*, vol. 14, no. 24, p. 11668, 2024.